

Chapter 3

Theory of Errors and Triangulation Adjustment



INTRODUCTION

In surveying measurements, there are all possibilities for errors to creep in, irrespective of the precautions and precision exercised. This necessitates a proper control, assessment and distribution of errors. Error, mistake and discrepancy are few terms frequently encountered in surveying measurements. True error may be defined as the difference in measured and true value of a quantity. But, since the true value of a measured quantity is not known, true error of the measurement is never known. However, when a number of measurements are to fulfill a known condition, true error can be obtained. For example, the theoretical sum of exterior included angles in a traverse of N sides is $(2N + 4) \times 90^\circ$. Here, the error in the sum of observed angles is known, but the error in individual angles is not known.

The observed difference between two like measurements each of which may have an error, of a quantity, is termed as discrepancy. A discrepancy should not be regarded as an error. A small discrepancy between two measurements does not reflect that error is small. Whereas, a large discrepancy indicates a mistake in the observations.

Mistakes are the errors arising from carelessness, inexperience, poor judgment and confusion of the observer.

The basis of a sound survey procedure depends upon the accuracy of field observations, and for an intelligent analysis of field data it is necessary to apply theory of errors. Some important aspects of the study of errors are nature of error, its assessment, i.e. how it propagates, and its distribution. The scope of the chapter is a combination of reduction and adjustment of field measurements to provide most probable and consistent values of the field observations (data).

Nature of Error

Errors in a measurement may be positive if the measurement is too large, or negative if the same is too small as compared to its true value. Errors are classified as systematic errors and accidental errors.

Systematic errors These are the errors which occur from well-understood causes and can be reduced by adopting suitable methods. For example, the error due to sag of a tape supported at its ends can be calculated and subtracted from each measurement. However, the tape can be supported throughout its length at short intervals and the sag error may be reduced to a negligible quantity. It always has the same magnitude

and sign so long as the conditions remain the same, and such an error is called constant *systematic error*. Whereas, if the conditions change, the magnitude of the error changes and is known as *variable systematic error*. A systematic error follows a definite mathematical or physical law and, therefore, a correction can always be determined and applied. These errors are also known as 'cumulative errors'.

Accidental errors These are the errors due to a combination of causes and are beyond the control of the surveyor. These can be positive or negative. Erroneous calibration of a chain is an example of an accidental error.

There is in reality no fixed boundary between the accidental and systematic errors. Every accidental error has some cause, and if the causes were perfectly understood and the amount and sign could be determined, it would cause an accidental error, but would be classed as systematic. On the other hand, a constant or systematic error may be brought into the accidental class wholly or partially by varying the conditions, instruments, etc. such that the sign of the errors is frequently reserved.

DEFINITIONS

3.1

3.1.1 Observed Quantities

An observed quantity may be either an independent quantity or a conditioned quantity. An *independent quantity* is one, whose value is independent of the influence of the value of any other quantity. The examples include values of R.L. of a B.M. and length of base line. On the other hand, the value of a *conditioned quantity* depends on the value of one or more quantities. The angles of a polygon of N sides are conditioned quantities, since the theoretical sum of the angles should be $(2N + 4) \times 90^\circ$.

3.1.2 Value of a Quantity

The value of a quantity may be the true value, the observed value, or the most probable value. A *true value* is defined as the value absolutely free from all errors, whereas the *observed value* is obtained as a result of an observation, after applying corrections for all the known errors. When many observations are made for the same quantity, the *most probable value* is the value which is more likely to be the true value. The difference between the most probable value and the observed value of a quantity is known as *residual error*.

3.1.3 Weight

The weight of an observation is measure of its relative trustworthiness. For a series of observations made under varying conditions (personal, instrumental or atmospheric), a uniform degree of reliability in the results would not be expected. Such observations are called *unequal observations*. Obviously, to derive the ultimate value of the quantity measured, it is not enough in this case to simply take the arithmetic mean, but the quality of each measurement will have to be taken into account.

The quality of each measurement result is evaluated by means of a number known as *weight*. The more reliable the result, the higher its weight. The weight thus expresses the amount of confidence enjoyed by this result as compared to the other results. The weights are established either by estimation or in accordance

with the measurement conditions. Since corresponding to the definite conditions of measurement there is a definite mean square error, it is most natural to establish the weight of the measurements with reference to the mean square error. The weights of the measurement results are assumed to be inversely proportional to the squares of their respective mean square error.

The following rules may be observed while assigning weights to the observations made:

1. The weight of an angle varies directly as the number of observations made on the angle.
2. For an angle repeated a great number of times, the weight is inversely proportional to the square of the probable error.
3. For a line of levels, the weight varies inversely as the length of the route.

Let w be the weight of the measurement, W the weight of the arithmetic mean of n such measurements, and m the mean square error. Then

$$\frac{W}{w} = \frac{m^2}{(m/\sqrt{n})^2} = n$$

Hence, the weight of an arithmetic mean is n times superior to that of the measurement results.

Numerical values assigned as the weights of a series of observations are simply ratios indicating the relative precision of the observations.

LAWS OF WEIGHT 3.2

1. The weight of the arithmetic mean of observations of individual weight is equal to the number of observations.

From the equation $\frac{W}{w} = n$, if $w = 1$, then $W = n$

2. The weight of the weighted arithmetic mean of observations of unit weight is equal to the sum of the individual weights.

Let angle A be measured three times by the repetition method

$\angle A$	Weight
48°20'8"	3
48°20'10"	2
48°20'12"	3

Sum of the weights = $3 + 2 + 3 = 8$

Weighted arithmetic mean

$$= 48^\circ 20' + \frac{1}{8} (8'' \times 3 + 10'' \times 2 + 12'' \times 3)$$

$$= 48^\circ 20' 10''$$

Weight of the weighted arithmetic mean $48^\circ 20' 10''$ is 8.

3. If two or more quantities are added algebraically, the weight of the result is equal to the reciprocal of the sum of the reciprocals of the individual weights.

For example,

Angle	Weight
$\alpha = 46^\circ 30' 30''$	3
$\beta = 24^\circ 42' 20''$	2

$$\text{Weight of } \alpha + \beta (71^\circ 12' 50'') = \frac{1}{\frac{1}{3} + \frac{1}{2}} = \frac{6}{5}$$

$$\text{Weight of } \alpha - \beta (21^\circ 28' 10'') = \frac{1}{\frac{1}{3} + \frac{1}{2}} = \frac{6}{5}$$

4. If a quantity is multiplied by a factor, the weight of the product is equal to the weight of that quantity divided by the square of the factor.

For example, if angle $\alpha = 32^\circ 12' 12''$ and weight = 2, then

$$\text{weight of } 3\alpha (96^\circ 36' 36'') = \frac{2}{3^2} = \frac{2}{9}$$

5. If a quantity is divided by a factor, the weight of the result equals the weight of that quantity multiplied by the square of that factor.

For example, if angle $\alpha = 41^\circ 31' 15''$ and weight = 2, then

$$\text{weight of } \frac{\alpha}{3} (13^\circ 50' 45'') = 2 \times 3^2 = 18$$

6. If an equation is multiplied by its own weight, the weight of the resulting equation is equal to the reciprocal of the weight of the equation.

For example, let angle $(A + B) = 189^\circ 40' 20''$ and weight of angle $(A + B) = 4/9$, then

$$\text{weight of } \frac{4}{9} (A + B), \text{ i.e. of } 84^\circ 17' 55.55'' = \frac{9}{4}$$

7. If an equation is added to or subtracted from a constant or if all the signs of the equation are changed, the weight remains unchanged.

For example, if $\angle A + \angle B = 78^\circ 45' 22''$ and its weight = 3 then,

$$\text{weight of } 180^\circ - (\angle A + \angle B), \text{ i.e. of } 101^\circ 14' 38'' = 3$$

Example 3.1

An angle A was observed by the method of repetition by three different observers and the values were as below.

Observer No.	$\angle A$	Number of measurements
1	40°20'	2
2	40°30'	3
3	40°45'	4

Find the most probable value of the angle.

Solution

The most probable value of the angle is equal to its weighted arithmetic mean.

$$40^{\circ}20' \times 2 = 80^{\circ}40'$$

$$40^{\circ}30' \times 3 = 121^{\circ}30'$$

$$40^{\circ}45' \times 4 = 163^{\circ}00'$$

$$\text{Sum} = 365^{\circ}10'$$

$$\Sigma \text{weight} = 2 + 3 + 4 = 9$$

$$\text{weighted arithmetic mean} = \frac{365^{\circ}10'}{9} = 40^{\circ}34'26.67''$$

The most probable value of angle A is $40^{\circ}34'26.67''$.

LAWS OF ACCIDENTAL ERRORS

3.3

A close examination of a set of observations reveals that the accidental errors follow the law of probability and it is because of this accidental errors are also known as *probable errors* or *standard errors*. This law defines the occurrence of errors and when expressed in the form of an equation, it can be used to compute the probable value or the probable precision of a quantity. In a series of observations, the probable error is a value such that the number of errors numerically greater than it is the same as that those less than it. It is always written after the observed quantity with the plus and minus sign, e.g. $25^{\circ}42'30'' \pm 3.16''$. The theory of a probability describes this with the help of a probability curve (Fig. 3.1). This curve forms the basis for the mathematical derivation of the theory of errors. Probable error serves two important purposes: (1) as a measure of the precision of any series of observations, and (2) as a means of assigning weights to two or more quantities, and thus to find the weighted mean or the most probable value of each.

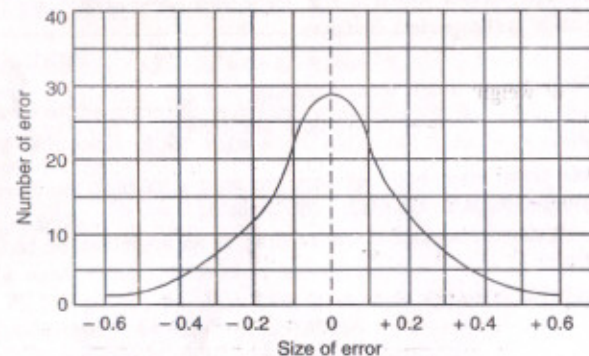


Fig. 3.1 Probability curve

It should be noted that the probable error is never applied as a correction to the mean value of a series. This is evidently impossible because the probable error carries the double sign, plus or minus. The mean value still remains the most probable value of the quantity being measured, and the only way in which its precision can be increased is (1) by increasing the number of observations in the field, or (2) by more refined and careful field methods. The following cases may arise.

3.3.1 Direct Observations of Equal Weight

Probable error of a single observation,

$$E_s = 0.6745 \sqrt{\frac{\Sigma v^2}{n-1}}$$

also

$$E_s = 0.845 \bar{v} \quad (\text{approx.})$$

Probable error of an average (arithmetic mean),

$$E_m = \frac{E_s}{\sqrt{n}}$$

Probable error of a sum of measurements,

$$E_{sm} = \sqrt{E_1^2 + E_2^2 + E_3^2 + \dots + E_n^2}$$

where

n = number of observations in the series

v = difference between any single observation and the mean of the series

$E_1, E_2, \dots, E_n$ = probable errors of several observations which sum up in a measurement

3.3.2 Direct Observations of Unequal Weight

Probable error of a single observation of unit weight,

$$E_{su} = 0.6745 \sqrt{\frac{\Sigma (wv^2)}{n-1}}$$

Probable error of an observation of weight w ,

$$E_{suw} = \frac{E_{su}}{\sqrt{w}}$$

Probable error of weighted arithmetic mean,

$$E_{smw} = \frac{E_s}{\sqrt{\Sigma w}}$$

where w = weight of an observation

3.3.3 Indirect Observations on Independent Quantities

Probable error of an observation of unit weight,

$$E_{si} = 0.6745 \sqrt{\frac{\Sigma ww^2}{n-q}}$$

Probable error of an observation of weight w ,

$$E_{siw} = \frac{E_{si}}{\sqrt{w}}$$

where q = number of unknown quantities.

3.3.4 Indirect Observations Involving Conditional Equations

Probable error of an observation of unit weight,

probable error from both sources will be the square root of the sum of squares of the errors from the individual sources:

$$(a) \sqrt{(\sqrt{2}\alpha)^2 + (\sqrt{2}\beta)^2} = \sqrt{2\alpha^2 + 2\beta^2} = \sqrt{2 \times 6^2 + 2 \times 6^2} = 12''$$

$$(b) \sqrt{(\alpha)^2 + (\sqrt{2}\beta)^2} = \sqrt{\alpha^2 + 2\beta^2} = \sqrt{6^2 + 2 \times 6^2} = 10.4''$$

$$(c) \sqrt{(\alpha/\sqrt{2})^2 + (\beta)^2} = \sqrt{\alpha^2/2 + \beta^2} = \sqrt{6^2/2 + 6^2} = 7.3''$$

Example 3.4 The following observations were recorded for an angle under identical conditions:

162°20'00"	162°21'20"	162°21'40"
162°20'40"	162°19'40"	162°21'20"

Calculate the (i) probable error of a single observation, (ii) probable error of the mean, and (iii) most probable value of the angles.

Solution

Value	Mean	v	v^2
162°20'00"		- 46.67"	36'18.08"
162°21'20"		+ 33.34"	18'31.56"
162°21'40"	162°20'46.66"	+ 53.33"	47'24.08"
162°20'40"		- 6.67"	44.49"
162°19'40"		- 1'6.67"	744.88"
162°21'20"		+ 33.34"	18'31.56"
		$\Sigma v = 0,$	$\Sigma v^2 = 3^\circ 15' 34.57''$

$$E_s = \pm 0.6745 \sqrt{3^\circ 15' 34.57'' / (6-1)} = 32.676'$$

Probable error of the mean $E_m = E_s / \sqrt{n} = 32.676' / \sqrt{6} = 13.34'$

Most probable value of the angle = $162^\circ 20' 46.66'' \pm 13.34' = 162^\circ 34' 7.06''$ or $162^\circ 7' 26.26''$.

Example 3.5 The most probable value of an angle A was found to be $44^\circ 32' 40'' \pm 2''$.

Calculate the most probable value of its supplement.

Solution We know, $A + A' = 180^\circ$

Hence, $A' = 180^\circ - 44^\circ 32' 40'' = 135^\circ 27' 20''$

Since the probable error of A' = the probable error of A,
the most probable value of $A' = 135^\circ 27' 20'' \pm 2''$

Example 3.6 The most probable values of two angles A and B are as given below:

$$\angle A = 47^\circ 32' 30'' \pm 2''$$

$$\angle B = 24^\circ 16' 40'' \pm 3''$$

Find the probable error of the angle (A + B).

Solution Let sum of the two angles be C,

$$\angle C = \angle A + \angle B = 47^\circ 32' 30'' + 24^\circ 16' 40'' = 71^\circ 49' 10''$$

$$e_C = \sqrt{e_A^2 + e_B^2} = \sqrt{2^2 + 3^2} = \pm 3.605''$$

Hence,

$$\angle C = 71^\circ 49' 10'' \pm 3.605''$$

Example 3.7

Calculate the most probable value and the most probable error of the area of a circle of radius 12.25 ± 0.03 m.

Solution Area, $A = \pi r^2 = \pi (12.25)^2 = 471.4353 \text{ m}^2$

$$\frac{dA}{dr} = 2\pi r$$

$$e_r = \pm 0.03$$

$$e_A = e_r \times \frac{dA}{dr} = \pm 0.03 \times 2 \times \pi \times 12.25 = \pm 2.309$$

Hence, the most probable value of the area = $471.4353 \pm 2.309 \text{ m}^2$

Example 3.8

The radius of a circle is 21 m with a probable error ± 0.21 . What will be the probable error of its circumference?

Solution Circumference, $C = 2\pi r$

$$= 2 \times \pi \times 21 = 132 \text{ m}$$

$$e_C = 2\pi e_r$$

$$= 2 \times \pi \times 0.21 = \pm 1.32$$

The probable error of the circumference is ± 1.32

Alternatively

$$C = 2\pi r$$

$$\frac{dC}{dr} = 2\pi$$

$$e_C = e_r \times \frac{dC}{dr}$$

$$= 0.21 \times 2\pi = \pm 1.32$$

Example 3.9

Following observations were recorded for horizontal angles of a triangle

$$\angle A = 20^\circ 10' \pm 0.2$$

$$\angle B = 100^\circ 40' \pm 0.1$$

$$\angle C = 59^\circ 10' \pm 0.2$$

Determine the probable error of its summation.

Solution Sum of angles, $S = \angle A + \angle B + \angle C$

Let e_A , e_B , and e_C be the probable errors of angles A, B, and C.

$$\text{Then, } e_S = \sqrt{e_A^2 + e_B^2 + e_C^2}$$

$$= \sqrt{0.2^2 + 0.1^2 + 0.2^2} = \pm 0.3$$

The probable error of summation is ± 0.3

Example 3.10

Calculate the most probable value and the probable error of the area of a rectangle whose sides are as follows.

$$\text{Side } a = 100 \pm 0.02 \text{ m}$$

$$\text{side } b = 150 \pm 0.01 \text{ m}$$

Considering errors

$$\varepsilon_1^2 + \varepsilon_2^2 + \dots + \varepsilon_n^2 = \frac{1}{2h_1^2} + \frac{1}{2h_2^2} + \dots + \frac{1}{2h_n^2}$$

Then,
$$\varepsilon^2 = \frac{1}{2h^2}$$

As h increases, $1/2h^2$ decreases, thus maximum accuracy is achieved when

$$\varepsilon^2 = 1/(2h^2) \text{ is minimum}$$

The main objects of the method of least squares are (a) to determine the best values which can possibly be obtained from a given set of measurements, (b) to determine the degree of dependence which can be placed upon these values, (c) to enable us to trace to their sources the various errors affecting the measurements, and consequently, (d) to increase the accuracy of the result by a proper modification of the methods and the instruments used.

It should be kept in mind that the method of least squares deals with accidental errors only and can tell us nothing about the constant or systematic errors which may affect the results of observations. A computation of the most probable value improves the result with respect to the accidental errors, but leaves the more serious form of error untouched. Multiplying observations and adjusting them for the purpose of removing small accidental errors, and at the same time failing to remove a large constant error may defeat the very purpose of the exercise.

Though the method of least squares may not show directly the presence of constant errors, yet a study of the precision of results, and a knowledge of the law governing the behaviour of accidental errors, may enable the observer to at least decide whether it is probably present. Consequently, the observer can modify his methods of observing so as to reduce the effect of such errors. Variations in the result, which are greater than the error of observation shown by precision measures, is likely to mean that systematic error is present. This tracing of errors to their sources, and the consequent modification of instruments and methods, may constitute the most important application of the method of least squares.

Example 3.15

The following observations were made in the rating tests of a current meter

Number of revolutions (N)	Velocity of flow (m/s)
0.3	0.50
1.0	1.50
1.5	2.30
2.1	2.78

The rating formula is $V = aN + b$

Find the constants a and b using the theory of least squares.

Solution The observation equations are

$$0.50 = 0.3 a + b$$

$$1.50 = 1.0 a + b$$

$$2.30 = 1.5 a + b$$

$$2.78 = 2.1 a + b$$

If a and b are the most probable values, then the error of observations are

$$0.50 - 0.3 a - b$$

$$1.50 - 1.0 a - b$$

$$2.30 - 1.5 a - b$$

$$2.78 - 2.1 a - b$$

By the theory of least squares

$$(0.5 - 0.3 a - b)^2 + (1.5 - 1.0 a - b)^2 + (2.3 - 1.5 a - b)^2 + (2.78 - 2.1 a - b)^2 = \text{minimum} \quad (1)$$

For normal equation in a , differentiating Eq. (1) w.r.t a , we get

$$2(-0.3)(0.5 - 0.3 a - b) + 2(-1.0)(1.5 - 1.0 a - b) + 2(-1.5)(2.3 - 1.5 a - b) + 2(-2.1)(2.78 - 2.1 a - b) = 0$$

$$\text{or } 10.938 - 7.75 a - 4.9 b = 0 \quad (2)$$

Now, differentiating Eq. (1) w.r.t b to get normal equation in b , we get

$$2(-1)(0.5 - 0.3 a - b) + 2(-1)(1.5 - 1.0 a - b) + 2(-1)(2.3 - 1.5 a - b) + 2(-1)(2.78 - 2.1 a - b) = 0$$

$$\text{or } 7.08 - 4.9 a - 4b = 0 \quad (3)$$

Eqs. (2) and (3) are simultaneous. These may be solved for a and b .

$$a = 1.296, \quad b = 0.182$$

$$\text{Hence } V = 1.296 N + 0.182$$

Example 3.16

The telescope of a theodolite is fitted with stadia wires. It is required to find the most probable values of the constants C and K of tachometer. The staff was kept vertical at three points in the field and with line of sight horizontal the staff intercepts observed were as follows

Distance of staff from tachometer	Staff intercept S (m)
D (m)	
150	1.495
200	2.000
250	2.505

Solution The distance equation is

$$D = KS + C$$

The observation equations are

$$150 = 1.495 K + C$$

$$200 = 2.000 K + C$$

$$250 = 2.505 K + C$$

If K and C are the most probable values, then the error of observations are:

$$150 - 1.495 K - C$$

$$200 - 2.000 K - C$$

$$250 - 2.505 K - C$$

By the theory of least squares

$$(150 - 1.495 K - C)^2 + (200 - 2.000 K - C)^2 + (250 - 2.505 K - C)^2 = \text{minimum} \quad (1)$$

For normal equation in K ,

differentiating Eq. (1) w.r.t. K , we get

$$2(-1.495)(150 - 1.495 K - C) + 2(-2.0)(200 - 2.00 K - C) + 2(-2.505)(250 - 2.505 K - C) = 0$$

$$\text{or} \quad 1250.50 - 12.51 K - 6C = 0$$

$$\text{or} \quad 208.41667 - 2.085 K - C = 0 \quad (2)$$

For normal equation in C ,

differentiating Eq. (1) w.r.t. C , we get

$$2(-1.0)(150 - 1.495 K - C) + 2(-1.0)(200 - 2.0 K - C) + 2(-1.0)(250 - 2.505 K - C) = 0$$

$$600 - 6K - 3C = 0$$

$$200 - 2K - C = 0 \quad (3)$$

Equations (2) and (3) are simultaneous. These may be solved for K and C .

$$K = 99.0196$$

$$C = 1.9608$$

$\therefore$ The distance equation is:

$$D = 99.0196 S + 1.9608 \quad \text{Ans.}$$

Example 3.17

The following levels along the centre line of a road were obtained at chainages of 100 m intervals under difficult conditions.

Chainage (m)	200	300	400	500	600
Level (m)	102.25	105.29	108.20	111.28	114.22

Assuming the grade to be a straight line, compute the most probable values of the level at chainage 100 m, and of the gradient.

Solution Level at any point will be

$$h = aC + b \quad (1)$$

where

C = chainage of the point

a, b = constants

The observation equations are given by

$$102.25 = 200 a + b$$

$$105.29 = 300 a + b$$

$$108.20 = 400 a + b$$

$$111.28 = 500 a + b$$

$$114.22 = 600 a + b$$

If a and b are the most probable values, then the error of observations are

$$102.25 - 200 a - b$$

$$105.29 - 300 a - b$$

$$108.20 - 400 a - b$$

$$111.28 - 500 a - b$$

$$114.22 - 600 a - b$$

By the theory of least squares

$$(102.25 - 200 a - b)^2 + (105.29 - 300 a - b)^2 + (108.20 - 400 a - b)^2 + (111.28 - 500 a - b)^2 + (114.22 - 600 a - b)^2 = \text{minimum} \quad (2)$$

For normal equation in a , differentiating Eq. (2) w.r.t. a , we get

$$2(-200)(102.25 - 200 a - b) + 2(-300)(105.29 - 300 a - b) + 2(-400)(108.20 - 400 a - b) + 2(-500)(111.28 - 500 a - b) + 2(-600)(114.22 - 600 a - b) = 0$$

$$\text{or} \quad 900\,000 a + 2000 b - 219\,489 = 0 \quad (3)$$

Now differentiating Eq. (2) w.r.t. b to get normal equation in b ,

$$2(-1)(102.25 - 200 a - b) + 2(-1)(105.29 - 300 a - b) + 2(-1)(108.20 - 400 a - b) + 2(-1)(111.28 - 500 a - b) + 2(-1)(114.22 - 600 a - b) = 0$$

$$\text{or} \quad 2000 a + 5 b - 541.24 = 0 \quad (4)$$

Equations (3) and (4) are simultaneous. These may be solved for a and b .

$$a = 0.02993 \text{ m} \quad \text{and} \quad b = 95.276 \text{ m}$$

Eq. (1) can be rewritten as:

$$h = 0.02993 C + 95.276$$

Now, the most probable level at chainage 100 m is

$$= 0.02993 \times 100 + 95.276 = 99.27 \text{ m}$$

$$\text{The most probable grade} = \frac{2.993}{100} = 1 \text{ in } 33.41$$

DISTRIBUTION OF ERROR TO THE FIELD MEASUREMENT

3.5

It is customary to check the observations made in the field for the closing error, if any. The closing error is then distributed to the various observed quantities as per the following rules.

1. The correction to be applied to an observation is inversely proportional to the weight of the observation.
2. The correction to be applied to an observation is directly proportional to the square of the probable error.
3. The correction to be applied to the line of levels is proportional to the length of the route.

Example 3.18

On a station O , three horizontal angles, closing the horizon, were measured as follows:

$$\angle A = 34^\circ 10' 20'' \pm 3'', \angle B = 176^\circ 40' 32'' \pm 4'', \angle C = 149^\circ 09' 04'' \pm 5''$$

Calculate the corrected angles.

$$\begin{aligned}\text{Sum of angles} &= 34^\circ 10' 20'' + 176^\circ 40' 32'' + 149^\circ 09' 04'' \\ &= 359^\circ 59' 56''\end{aligned}$$

$$\text{Discrepancy} = 360^\circ - 359^\circ 59' 56'' = 04''$$

This error of $04''$ is to be distributed in proportion to the square of the probable error in each angle. Let c_1 , c_2 and c_3 be the corrections to be applied to the angles A , B and C , respectively.

$$c_1 : c_2 : c_3 = (3)^2 : (4)^2 : (5)^2 = 9 : 16 : 25$$

$$\text{Hence, } c_2 = (16/9) c_1 \text{ and } c_3 = (25/9) c_1$$

$$\text{Also, } c_1 + c_2 + c_3 = 4''$$

$$\text{or } c_1 + (16/9) c_1 + (25/9) c_1 = 4''$$

$$\text{or } (50/9) c_1 = 4''$$

$$\text{or } c_1 = (4 \times 9)/50 = 0.72''$$

$$\therefore c_2 = (16/9) \times 0.72 = 1.28''$$

$$\text{and } c_3 = (25/9) \times 0.72 = 2.00''$$

Check,

$$c_1 + c_2 + c_3 = 0.72'' + 1.28'' + 2.00'' = 4''$$

Hence, the corrected angles are:

$$\angle A = 34^\circ 10' 20'' + 0.72'' = 34^\circ 10' 20.72''$$

$$\angle B = 176^\circ 40' 32'' + 1.28'' = 176^\circ 40' 33.28''$$

$$\angle C = 149^\circ 09' 04'' + 2.00'' = 149^\circ 09' 06''$$

Example 3.19 Neglecting spherical excess, adjust the angles of a triangle of which the observed values are

$$\angle A \quad 48^\circ 18' 22'' \quad \text{Weight} = 3$$

$$\angle B \quad 76^\circ 32' 47.2'' \quad \text{Weight} = 1$$

$$\angle C \quad 55^\circ 08' 53.8'' \quad \text{Weight} = 3$$

Solution The sum of the angles of a triangle should be 180° .

$$\angle A + \angle B + \angle C$$

$$= 48^\circ 18' 22'' + 76^\circ 32' 47.2'' + 55^\circ 08' 53.8'' = 180^\circ 0' 3''$$

There is an error of $3''$ which should be distributed to the various angles inversely as their weights. Let C_A , C_B and C_C be the respective corrections to the angles A , B and C .

$$\text{Then, } C_A : C_B : C_C = \frac{1}{3} : \frac{1}{1} : \frac{1}{3} \quad (1)$$

$$\text{and } C_A + C_B + C_C = -3''$$

From Eq. (1)

$$C_A = C_A$$

$$C_B = 3C_A$$

$$C_C = C_A$$

Substituting these values in Eq. (2)

$$C_A + 3C_A + C_A = -3''$$

$$\text{Hence, } C_A = -0.6''$$

$$C_B = -3 \times 0.6 = -1.8''$$

$$C_C = -0.6''$$

Handwritten notes:

$$\frac{C_A}{C_B} = \frac{1/3}{1} = \frac{1}{3}$$

$$\frac{C_A}{C_C} = \frac{1/3}{1/3} = 1$$

$$C_A = \frac{1}{3}, C_B = 1, C_C = \frac{1}{3}$$

$$C_A = -0.6, C_B = -1.8, C_C = -0.6$$

Corrected values of the angles are

$$\angle A = 48^\circ 18' 22'' - 0.6'' = 48^\circ 18' 21.4''$$

$$\angle B = 76^\circ 32' 47.2'' - 1.8'' = 76^\circ 32' 45.4''$$

$$\angle C = 55^\circ 08' 53.8'' - 0.6'' = 55^\circ 08' 53.2''$$

Check:

$$\text{Total} = 180^\circ 00' 00''$$

NORMAL EQUATIONS 3.6

An 'observation equation' is the symbolic equality of a quantity and its observed value, e.g. $\angle A = 145^\circ 32' 20''$, or $\angle B + \angle C = 67^\circ 35' 17''$, the former value is observed by direct single measurement and the latter one is observed indirectly when simultaneous observations or conditions are involved.

Consider a measurement of angles A , B and C closing the horizon at a station O . The rigorous geometrical condition to be fulfilled is $\angle A + \angle B + \angle C = 180^\circ = -k$ (say).

Now, if the angles are measured with the same precision, the error e is $A + B + C + k$ and the most probable value of each of the angles will be obtained by applying $(1/3)e$ to each of the observed values. However, if one angle is measured directly and the others indirectly by subtraction from the observed values of two or three angles together (reiteration method of measuring horizontal angles), or in a way that the conditions of measurements are varied, the error equation takes the form

$$e = aA + bB + cC + k$$

and if the measurements are repeated giving separate observations A_1, B_1, C_1 and A_2, B_2, C_2 , etc., the errors will be

$$e_1 = aA_1 + bB_1 + cC_1 + k$$

$$e_2 = aA_2 + bB_2 + cC_2 + k$$

From the theory of errors, $\Sigma(e)^2 = \Sigma(aA + bB + cC + k)^2$ should be minimum.

To minimize this expression, differentiate it with respect to A , B and C and equating each result to zero. We get,

$$\Sigma a(aA + bB + cC + k) = 0$$

$$\Sigma b(aA + bB + cC + k) = 0$$

$$\Sigma c(aA + bB + cC + k) = 0$$

This shows that the fundamental equation when multiplied by coefficients of $\angle A$, $\angle B$ and $\angle C$ will provide the equations, the simultaneous solution of which will lead to the most probable values of $\angle A$, $\angle B$ and $\angle C$. Such equations are known as 'normal equations'.

If the three angles in the above example are measured with different precision and are accorded weights w_A , w_B and w_C , the error equations will be

$$\Sigma w_A a(aA + bB + cC + k) = 0$$

$$\Sigma w_B b(aA + bB + cC + k) = 0$$

$$\Sigma w_C c(aA + bB + cC + k) = 0$$

The solution of the normal equations may be obtained by the direct method or by the method of differences.

3.6.1 Direct Method

In the direct method each side of the observation equation is multiplied by the weight of the observation. The weighted observations containing that quantity are then multiplied by their respective coefficients to yield normal equations. These equations are then solved simultaneously to determine the individual and the most probable values of the quantities.

3.6.2 Method of Differences

In the beginning, the reported single observations are assumed to be the most probable values, and for the remaining quantities, the most probable value is obtained by subtracting it from two or three variable equations. Let the discrepancy be $k_1, k_2, \dots$, etc. The observation equations are then formulated in terms of $k_1, k_2, \dots$, etc. to express the discrepancies between the observed results and those given by the assumed values, always subtracting the latter from the former. The equations thus obtained are multiplied by the respective weights, and the normal equations are formed by multiplying each side by the coefficients of $k_1, k_2, \dots$, etc. These simultaneous equations are then solved for $k_1, k_2, \dots$, etc. and are added algebraically to the assumed values of the quantities.

The arithmetic work is reduced in this method and hence it is particularly advantageous when the number of conditions are more.

Example 3.20 Angles were measured on a station and the observations were recorded as follows.

Angle	Value	Weight
A	45°30'10"	2
B	40°20'20"	3
A + B	85°50'10"	1

Find the most probable values of the angles A and B.

Solution

(i) Direct method

$$\begin{aligned}\angle A &= 45^\circ 30' 10'' & \text{Weight} &= 2 \\ \angle B &= 40^\circ 20' 20'' & \text{Weight} &= 3 \\ \angle A + \angle B &= 85^\circ 50' 10'' & \text{Weight} &= 1\end{aligned}$$

Multiply both the sides of the weighted observations by algebraic coefficients of A, thus forming the normal equations in A.

$$\begin{aligned}2A &= 91^\circ 00' 20'' \\ A + B &= 85^\circ 50' 10'' \\ \hline 3A + B &= 176^\circ 50' 30''\end{aligned}\quad (1)$$

Likewise, the normal equation in B is formed

$$\begin{aligned}3B &= 121^\circ 01' 00'' \\ A + B &= 85^\circ 50' 10'' \\ \hline A + 4B &= 206^\circ 51' 10''\end{aligned}\quad (2)$$

Solve the above two simultaneous equations for the values of A and B.

Multiply Eq. (1) by 4 and subtract Eq. (2) from it to get

$$\begin{aligned}12A + 4B &= 707^\circ 22' 00'' \\ A + 4B &= 206^\circ 51' 10'' \\ \hline 11A &= 500^\circ 30' 50''\end{aligned}$$

$$\text{or } \angle A = 45^\circ 30' 4.55''$$

Likewise for angle B, multiply Eq. (2) by 3 and subtract Eq. (1) from it to get

$$\begin{aligned}3A + 12B &= 620^\circ 33' 30'' \\ 3A + B &= 176^\circ 50' 30'' \\ \hline 11B &= 443^\circ 43' 00''\end{aligned}$$

$$\text{or } \angle B = 40^\circ 20' 16.36''$$

The most probable value of $\angle A$ is $45^\circ 30' 4.55''$ and that of $\angle B$ is $40^\circ 20' 16.36''$.

(ii) Method of differences

Let k_1 and k_2 be the most probable corrections to angles A and B. Then,

$$\angle A = 45^\circ 30' 10'' + k_1 \quad (1)$$

$$\angle B = 40^\circ 20' 20'' + k_2 \quad (2)$$

Add Eqs. (1) and (2)

$$\angle A + \angle B = 85^\circ 50' 30'' + k_1 + k_2 \quad (3)$$

Subtract these equations from the observed equations

$$k_1 = 0 \quad \text{Weight} = 2$$

$$k_2 = 0 \quad \text{Weight} = 3$$

$$k_1 + k_2 = -20'' \quad \text{Weight} = 1$$

Normal equation for k_1 is

$$2k_1 = 0$$

$$k_1 + k_2 = -20''$$

$$3k_1 + k_2 = -20''$$

Normal equation for k_2 is

$$3k_2 = 0$$

$$k_1 + k_2 = -20''$$

$$k_1 + 4k_2 = -20''$$

From the two simultaneous Eqs. (4) and (5)

$$k_1 = -5.45'' \quad \text{and} \quad k_2 = -3.63''$$

Hence, the most probable value of angle A

$$= 45^\circ 30' 10'' - 5.45'' = 45^\circ 30' 4.55''$$

and the most probable value of angle B

$$= 40^\circ 20' 20'' - 3.63'' = 40^\circ 20' 16.37''$$

METHOD OF CORRELATES

3.7

Correlates are the unknown multiples or independent constants used for finding most probable values of unknowns.

Space based Platforms:-

- Synoptic view [Large area in single image]

- Less affected by atmospheric drag.

- Systematic & repetitive Coverage.

Polar-Orbiting Satellites

- 700-1500 km

Geo-Stationary Satellites

[36000 km]

Facilities
Management

Disasters.

Digital
Elevation

Model

Coastal Environment

Ocean resources

Water shed

Environment...

Street-network

based application

Land parcel based...

Natural resources....

Agriculture

Forestry

Land use & soils

Geology

Urban Land use

Water Resources

1) Remote Sensing..... [Definition.....]

2) Electro-Magnetic Energy [Light, Heat, Microwave]

3) Air-Crafts & Satellites---

4) Highly sophisticated sensors
[Camera, Multi-spectral scanner, Radar etc]

5) Photo-interpretation-----

6) Active Remote Sensing
Passive Remote Sensing

7) Remote Sensing observation platforms...

→ Air borne platforms

→ Space based platforms

Air-borne:-

- Aircraft

- Costly & Time Consuming..

Global Positioning System:-

- Navstar [US DoD]
- Global Navigation Satellite System

GPS Satellites



GPS receiver [Passive receiving]

unobstructed view of the sky...

Time reference ---- Atomic clocks.....

GPS operations depend on time reference...

GPS satellites → Indicates its location & Time...

Signals → Moving at the speed of the light...

Min four satellites' distances & time → it can calculate its position in three dimension.

24 operational satellites + no. of spares...

- All GPS satellites have atomic clocks... [Atomic physics... one second in 30 million years]
- Each satellite will last only for 10 years
- "Longitude, Latitude, Altitude, Time" [Business transactions....]
- NAVSTAR - Navigation System using Timing And Ranging..